

ORAC-NT

Hardware-Tested Low-Latency Gravitational Wave Trigger Architecture

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Executive Summary

ORAC-NT is an edge-compatible, hardware-validated real-time signal detection pipeline originally developed for gravitational wave astronomy. Operating live via NASA GCN Kafka, the system has been benchmarked through five validation phases on real LIGO O3 data, achieving ROC AUC=0.983, 100% detection efficiency above SNR=10, and zero false alarms over 7.96 hours of real detector noise. The core algorithm runs at 535 nanoseconds on STM32F401 embedded hardware — making it uniquely suited for space-based and edge sensing applications where traditional pipeline infrastructure is unavailable.

System Architecture

Component	Description	Status
Multi-template MF bank	BBH/BNS/NSBH chirp templates, 20-500Hz	■ Validated
Kurtosis + Chi ² veto	Signal/glitch discrimination	■ Complete separation
Time-slide background	Statistical FAR estimation	■ 5σ = SNR 5.93
H1+L1 coincidence	10ms window, physics-motivated	■ 6/8 confirmed
NASA GCN Kafka client	Live 8-topic monitoring	■ 30+ events archived
STM32F401 deployment	Hardware trigger validation	■ 535ns latency

Validation Results (Phases A–E)

Phase	Test	Result
A	5/5 canonical GW events (BNS/BBH/NSBH)	100% detection, FAR=0
B	10/10 event generalization + injection campaign	100% generalization, 65-80% injection recovery
C	H1+L1 coincidence + glitch rejection + latency	6/8 coincidence, AUC=0.638, 130ms CPU
D	Long-duration FAR + ablation + PSD robustness	FAR=0, pipeline stable
E1	Self-calibrated matched filter (GW170817)	Peak SNR=11.73, threshold=8
E2	Time-slide background estimation	5σ threshold = SNR 5.93
E3	Sensitivity curve	50% efficiency @ SNR≈8, 100% @ SNR≥10
E4	ROC curve analysis	AUC = 0.983
E5	Chi ² signal/glitch discriminator	Complete separation, zero overlap

Latency Profile

Stage	Software (CPU)	Hardware (STM32F401)
Whitening	15ms	—
SNR Compute	108ms	—

Trigger Find	8ms	—
Total Pipeline	130ms	535ns ← unique advantage
GCN Alert Ingest	< 2s	—

Space & Edge Computing Relevance

Application	ORAC-NT Capability	Relevance
LISA preprocessing (ESA, 2034)	535ns embedded trigger Edge-compatible architecture	Direct: space-based GW detector needs edge processing
CubeSat anomaly detection	STM32-validated Low power consumption	Direct: onboard real-time signal discrimination
Multi-messenger astronomy	Live NASA GCN connection 8 alert topics monitored	Direct: rapid response to astrophysical transients
Critical infrastructure monitoring	Seismic/acoustic adaptation Deterministic latency	Dual-use: same algorithm applies to Earth sensors

Development Roadmap (E6–E10)

Milestone	Description	Timeline
E6: Real O4 validation	Test on confirmed O4 public data	Q3 2026
E7: IFAR curve	Inverse FAR vs threshold characterization	Q3 2026
E8: Multi-detector SNR	Network SNR H1+L1+V1 combined	Q3 2026
E9: IR1 live monitoring	Real alerts during Oct-Nov 2026 observing run	Q4 2026
E10: LISA adaptation	Prototype space-frequency band pipeline	Q1 2027

Commercial Relevance

ORAC-NT addresses a gap that no current commercial product fills: a verified, edge-deployable real-time signal discrimination engine with sub-microsecond hardware latency and published scientific validation. The technology is relevant across three market segments: (1) space instrumentation — preprocessing for next-generation GW detectors; (2) defense and maritime sensing — deterministic anomaly detection for sonar, seismic, and structural monitoring; (3) scientific instrumentation — independent pipeline for multi-messenger astronomy. The core IP (calibration methodology, H-factor algorithm, χ^2 reweighting) is protected under a non-commercial open license with commercial licensing available.